Benchmark Comparison: AFFORD-Bench vs. LIBERO-PRO & LIBERO-X

Key Differences in Physical Affordances, Perceptual Modalities, Policy Universality, and Scale

RSS 2027 BENCHMARK

BENCHMARK DIMENSION Core Evaluation Axes	LIBERO-PRO Zhou et al., 2026 • Robustness	LIBERO-X Wang et al., 2026 • Language-X	★ AFFORD-Bench (Ours) Physics-Grounded Universal Benchmark
Contact Physics Friction & mass dynamics	Nominal (Fixed μ = 0.5)	Nominal (Fixed μ = 0.5)	✓ True Contact Pads (μ ∈ [0.12, 0.25] & Mass)
Perception Inputs Visual & geometric sensors	1× RGB (128 × 128)	2× RGB (256 × 256)	✓ Dual RGB (640×660 / 300×420) + 3D Depth
Policy Universality Policy & dataset formats	VLA Models Only	VLA Models Only	✓ Universal + LeRobot v2.0 & RLDS
Domain Scope Environment breadth & tasks	40 Eval Tasks (4 Axes)	600 Procedural Tasks	✓ 210 Defined Tasks (7 Domains)
Precision & Anti-Cheat Tolerance & impulse probes	Coarse / Kinematic Welds	Coarse / Kinematic Welds	✓ 2–2.5 mm Tol. + Reactive Impulse Probe
Verified Solvability Empirical sanity baselines	Unverified Null Baselines	Unverified Null Baselines	✓ 99.8% Nominal Oracle (0.0% Null/Rand)
Benchmark Scale Normalized data volume	40 Tasks • Eval Only (0 Demos)	600 Tasks • 2,520 VR Demos	★ 210 Tasks • 10,500 Demos Target